Evaluating Tail Parameter Estimation Methods for Extreme Value Distributions: Toward Confidence Intervals for Integrated Information

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***Abstract*—Integrated Information Theory (IIT) provides a mathematical framework for studying consciousness, but calculating integrated information can become computationally infeasible as the number of possible system partitions increases. This motivates the question of whether the minimum of a large collection of values can be estimated using only an observed subset. The goal of this study is to evaluate and compare different tail-parameter estimation methods used within an Extreme Value Theory (EVT) approach for estimating this minimum. A simulation study was performed using four distributions in the Weibull domain of attraction and multiple subsample proportions. Estimator performance was evaluated using confidence interval coverage, average confidence interval length, bias, variance, and an overall quality score. The true-α benchmark produced the highest overall quality score. Among methods where α was estimated, Grimshaw MLE produced the strongest overall performance, while GPD MLE and Zhang LME also performed well. These results identify promising tail-parameter estimation methods for further study of minimum estimation from incomplete observations.**

I. Introduction

Consciousness is one of the most difficult and important problems in science today. Humans have conscious experiences such as thoughts, emotions, sensations, memories, and awareness. Conscious entities experience a life that is unique to them, best described as "there is something it is like to be you". Neuroscience has made progress in explaining many functions of the brain, such as memory, language perception, and more, but explaining what in the brain produces consciousness still remains a challenge.

Before Integrated Information Theory (IIT), there existed many theories of consciousness, where it may exist, and how it is described. Most theories focused on external behaviours such as intelligence, or specific activity within the brain. The issue with these approaches is that researchers and philosophers never figured out how to explain the subjective experience of being conscious, which is where the "hard problem of consciousness" arises [1].

Integrated Information Theory (IIT) was developed as an attempt to study consciousness scientifically through the scope of math and neuroscience. It was first introduced by neuroscientist Giulio Tononi [4]. IIT argues that consciousness is not related to intelligence or behaviour; it comes from how information is integrated within a physical system [1], [4]. The theory states that a system becomes conscious when its components interact in such a way that the system forms a unified and irreducible system [9], [11]. IIT uses the quantity phi (φ), which represents the amount of integrated information in a system [3], [4]. The higher the value of φ, the more integrated the system is. Higher values of φ are proposed to correspond to a more complex conscious experience [1]. Rather than focusing on outward behaviours, Tononi approached the problem to find the fundamental properties that all conscious experiences share in common. He argued that consciousness is unified, informative, and irreducible, meaning we cannot separate the experience into multiple parts without losing the experience as a whole [9], [11]. Additionally, IIT proposes that consciousness arises from the way information is integrated within a physical system. This led to the use of φ, which measures the amount of integrated information within a physical system [3], [5].

Another reason why IIT is heavily studied is that it is a way to unify multiple fields of research into a single framework. Consciousness historically has been examined separately through philosophy, psychology, neuroscience, and computer science. Mixing all these core subjects can lead to many conflicting explanations and differences in methodology. IIT bridges the gap between disciplines by proposing that consciousness can be understood both mathematically and philosophically [1], [9]. This approach has made the theory influential in discussions surrounding the relationship of the brain, information processing, and subjective experience.

Researchers aim to study IIT as it offers a framework to change consciousness from a philosophical problem to a numerical problem [2], [3]. Other approaches to this problem rely on external behaviours to determine whether the system is aware at that moment. This is flawed as we cannot test other conscious entities using this approach. Another flaw with this methodology is that you would have to be over a base level of consciousness to be able to answer questions, and needing to answer the questions may trigger unwanted side effects, such as other parts of the brain firing.

This becomes an issue when studying subjects who may still possess some level of consciousness but are unable to communicate or respond to questions. For example, a patient who is unconscious or minimally responsive may not be able to provide a verbal or behavioural response, even if some level of conscious experience is still present. Additionally, asking a subject to respond introduces additional brain activity related to understanding the question, forming a response, and communicating that response. This can make it more difficult to separate the conscious experience being studied from the activity required to report that experience. IIT states that consciousness depends on the internal structure of the system rather than only its outward behaviour. This provides a different way of approaching the problem by focusing on the interactions within the system itself. However, reports and behavioural responses are still important when testing theories of consciousness experimentally [1], [11]. This is important as two systems could behave the same way but have two different levels of consciousness. By focusing on IIT, we attempt to provide an explanation as to why certain physical systems have consciousness and some may not.

The growing interest in IIT also reflects broader scientific efforts to understand the human brain. Despite major technological advancements in brain imaging and neuroscience, researchers still lack a complete explanation for how billions of neurons collectively produce awareness and subjective experience. IIT offers a potential framework for investigating this relationship by emphasizing the importance of connectivity and causal interaction within neural systems [3], [7]. Even if the theory is later modified or challenged, studying IIT contributes to the broader goal of developing testable scientific models of consciousness.

The study of IIT has important applications in neuroscience and medicine. Understanding consciousness could improve the ability to evaluate patients who are unconscious or minimally responsive due to brain injuries, anesthesia, or disorders such as comas and vegetative states [1]. Researchers hope that measuring integrated information may help determine whether some level of consciousness is still present in patients who are unable to communicate [1]. IIT also contributes to research involving sleep, dreaming, and altered states of consciousness by examining how changes in brain connectivity affect conscious experience. IIT has influenced modern discussions around neural complexity and information processing in the brain. Researchers who study the brain often try to understand how large-scale interactions between brain regions contribute to awareness and perception. IIT provides us with a theory to understand why highly interconnected systems may produce richer conscious experiences than systems that are limited or disconnected [1], [3].

IIT challenges many assumptions regarding consciousness. For example, the theory states that consciousness may exist in varying degrees rather than just being present or absent. Instead, different systems may have different levels of consciousness depending on how information is integrated within the system. This means that consciousness does not necessarily have to be viewed as something that is either completely present or completely absent. A system may have a lower level of consciousness, while another system may have a much higher level. In IIT, the value of integrated information, φ, is used to quantify this integration. A larger value of φ represents a greater amount of integrated information within the system. This implies that some animals, simple neural systems, or even certain AIs may have a small amount of conscious experience depending on the level of integrated information [1]. Although these ideas are debated, they encourage people to reconsider long-standing beliefs about consciousness. According to IIT, different conscious experiences correspond to different informational structures generated within the brain [9], [11]. IIT does not only ask why consciousness exists, but also why experiences feel different from one another, such as seeing colours, hearing sounds, or experiencing emotion.

Furthermore, IIT has become more relevant with the emergence of AI and machine learning. As computers and AI systems continue to mimic human behaviour, scientists and philosophers continue to debate whether AI systems are conscious. IIT provides a unique perspective on consciousness, as the theory states that intelligence does not necessarily correlate to consciousness [1]. If IIT is correct, an entity could in theory act like a human in everything it does, but without actually possessing subjective experience in its internal structure it may lack integrated information. This has important implications for the future of AI, as an artificial system could display highly intelligent behaviour without necessarily possessing consciousness. This has also led to important discussions regarding the ethical implications of creating a system that may possess consciousness.

As AIs continue to advance, understanding the conditions required to be conscious may influence how society treats artificial intelligence systems. IIT forces people to consider fundamental questions about identity, awareness, and what it truly means to be conscious and possess experience.

IIT provides us with a mathematical framework for studying consciousness [3], [9]. Applying the theory can become computationally exhausting because as the size of the system increases, we must evaluate all possible partitions of a system and find the partition that produces the minimum value. This minimum is important because φ is defined using the weakest link in the system, which corresponds to the partition that produces the smallest information loss. Therefore, calculating φ requires comparing the information loss across the possible partitions and identifying the minimum. As the number of elements in the system increases, the number of possible partitions also increases, making this process increasingly computationally expensive [10]. This raises the question posed in the study: rather than evaluating the complete set, can the minimum be estimated while knowing only a subset of the available values? Since this minimum-estimation procedure depends on the estimation of the lower-tail parameter α, this study focuses on comparing different methods for estimating α and determining which methods perform best.

II. Experiment Setup and Methodology

A. Method Overview

The overall goal of the method is to estimate the minimum of a large finite collection of values when only a subset of those values is observed. This problem is motivated by Integrated Information Theory (IIT), where the number of possible partitions of a system can grow rapidly as the size of the system increases. Evaluating every possible partition to obtain the absolute minimum becomes computationally exhausting.

The method uses the behavior of the lower tail of an observed subsample to construct a confidence interval for the minimum of the full sample. Since this procedure depends on the lower-tail parameter α, the estimation of α is an important part of the method. Therefore, the goal of this study is to evaluate and compare different tail-parameter estimation methods and determine how the choice of estimator affects the final confidence interval.

B. Ordered Observations and Tail Threshold

Since the method is used to estimate a minimum, the most important information exists in the observations closest to the lower endpoint. Therefore, only the first k order statistics are used when estimating the lower tail.

The value k controls the amount of information considered to belong to the tail. Choosing k introduces a bias-variance tradeoff in extreme value estimation. A small value of k focuses only on the most extreme observations and therefore tends to better model the tail, but using very few observations can result in a lot of variance.

Increasing k gives more observations and can reduce variance, but as k moves further away from the extreme, observations from the central portion of the distribution may be added into the model, increasing bias. For this reason, different values of k are considered, and an optimal k is selected for each estimator because different estimation methods may reach their best performance using different amounts of tail information.

C. Weibull to Frechet Tail Transformation

Many Extreme Value Theory (EVT) estimators are made for heavy upper tails, but the problem considered in this study looks at a finite lower endpoint. To make these estimators applicable, each candidate lower endpoint is used to transform the lower-tail problem into an upper-tail problem.

For a candidate endpoint w, the observations are transformed according to

$$Y_i = \frac{1}{X_i - w}.$$

As an observation X_i approaches w from above,

$$X_i \to w^+,$$

the transformed observation becomes increasingly large:

$$Y_i \to \infty.$$

Therefore, observations that are extreme in the lower tail of the original distribution become extreme observations in the upper tail of the transformed distribution. This transformation converts the Weibull lower-tail problem into a Frechet-type upper-tail problem, allowing existing tail estimators to be used to estimate the lower-tail parameter α.

D. Tail-Parameter Estimation

For every candidate endpoint and each value of k, the transformed observations are supplied to an EVT tail-parameter estimator. The estimator produces an estimate $\hat{\alpha}$.

The tail-parameter estimators considered in this study are GPD maximum likelihood estimation (GPD MLE) [16], Grimshaw maximum likelihood estimation (Grimshaw MLE) [17], the Hill estimator [13], the L-Moment estimator [15], the Moment estimator [14], maximum product of spacings for the GPD (MPS GPD) [18], and Zhang likelihood moment estimation (Zhang LME) [19]. These methods represent different approaches to estimating the tail parameter and allow their performance to be compared within the same minimum-estimation procedure.

The purpose of comparing multiple estimators is to determine how sensitive the final minimum confidence interval is to the method used to estimate α and to see whether different methods perform differently under different conditions. Each estimator is given the same subsample so that the methods can be compared under the same conditions.

E. Lower Tail Quantile Model

After estimating the tail parameter, the original lower-tail observations are modeled using the quantile relationship

$$\ell + ap^{1/\alpha},$$

where ℓ represents the lower endpoint, $a > 0$ is a scale parameter, and α controls the shape of the lower tail.

For the observed sample of size n, plotting probabilities are assigned to the first k order statistics using

$$\frac{i}{n+1}, \qquad i = 1, \ldots, k.$$

The lower order statistics are approximately represented by

$$X_{(i)} \approx \ell + a\left(\frac{i}{n+1}\right)^{1/\hat{\alpha}}.$$

Once $\hat{\alpha}$ has been given by the EVT estimator, the problem becomes linear, where ℓ and a can be estimated using the observed first k order statistics.

F. Estimation of the Full-Sample Minimum

After estimating the lower-tail distribution from the subsample, the fitted model can be used to predict the minimum of the complete sample. Since n observations have already been observed, there are $m - n$ unobserved values that could have a value smaller than the subsample minimum.

Because the observed subsample is contained within the complete sample, the full-sample minimum cannot be greater

than the smallest value already observed. The confidence interval therefore combines the behavior of the unobserved lower tail with the known minimum of the observed subsample.

G. Simulation Study

A simulation study is used to evaluate and compare the performance of the different tail-parameter estimation methods within this procedure. Using simulated data allows the true minimum and tail parameter to be known, while only a subset of the observations is made available to the estimation procedure.

For each repetition, a complete sample of $m = 10000$ independent and identically distributed observations is generated from a distribution F:

$$X_1, X_2, \ldots, X_m.$$

The actual minimum of the complete sample is recorded as $X_{(1)}$. A random subsample of size n is then selected without replacement from the complete sample, where

$$n = pm$$

and

$$p = [0.01, 0.05, 0.1].$$

The true minimum is not given to the tail estimator or confidence-interval calculation unless it is randomly selected as part of the subsample. It is kept only so that the resulting confidence interval can later be evaluated as either containing or failing to contain the actual minimum.

For each distribution, the simulation is repeated 500 times. During each repetition, a 95% confidence interval is constructed for the minimum of the complete sample.

H. Data Generating Distributions

The confidence interval method is designed to be valid for any distribution belonging to the Weibull domain of attraction. These distributions have a finite lower endpoint, making them appropriate for studying minimum estimation using EVT.

To evaluate the method in a controlled setting, the simulation considers four distributions: Uniform$(0, 1)$, Beta$(1, 2)$, Beta$(2, 1)$, and Power(1). Although these distributions have different overall shapes, their behaviour near the lower endpoint belongs to the Weibull domain of attraction.

I. Selection of k

The number of observations used to represent the lower tail is k. For each estimator, k is selected by searching every 10th value in the range

$$4 \leq k < n$$

for 15 repetitions. The best-performing value is then selected and used for the full 500 simulation repetitions.

Therefore, the experiment does not assign a single value of k to every estimator. Each estimator receives its own selected value because different estimation methods may perform best using different amounts of tail information.

J. Performance Metrics

The performance of each estimator is evaluated using both the behaviour of the confidence intervals and the accuracy of the estimated tail parameter.

The first metric is coverage. For repetition r, we define coverage as

$$I_r = \begin{cases} 1, & L_r \leq X_{(1:m),r} \leq U_r, \\ 0, & \text{otherwise.} \end{cases}$$

Across R repetitions, coverage is calculated as

$$C = \frac{1}{R} \sum_{r=1}^{R} I_r.$$

Because the confidence level is 95%, a good method should produce coverage as close to 95% as possible. Coverage below 95% shows that the interval fails to contain the true minimum often enough, while higher coverage may hint at overly conservative intervals.

Coverage alone is not enough, since very wide intervals may have high coverage while providing little precision. Therefore, the average confidence interval length is also recorded:

$$\bar{L} = \frac{1}{R} \sum_{r=1}^{R} (U_r - L_r).$$

Shorter intervals are preferable when coverage is similar because they provide more precise information about the location of the minimum.

The statistical behavior of the tail-parameter estimator is evaluated using bias and variance. Bias is estimated as

$$\text{Bias}(\hat{\alpha}) = \frac{1}{R} \sum_{r=1}^{R} (\hat{\alpha}_r - \alpha),$$

while the variability of the estimates is estimated as

$$\text{Var}(\hat{\alpha}).$$

Considering both quantities is important because an estimator may have low bias but high variability, or high bias and low variability.

Coverage and interval length are combined into the summary quality score

$$Q = \frac{1}{\bar{L}\,(1 + 50|C - 0.95|)}.$$

The term

$$|C - 0.95|$$

measures the deviation from the coverage target and penalizes both undercoverage and overcoverage. The factor of 50 is the strength of this penalty, while the interval-length component rewards greater precision. Higher values of Q indicate better overall confidence-interval performance. The quality score is used alongside the individual performance metrics rather than replacing them.

III. Results

A. Coverage vs subsample proportion

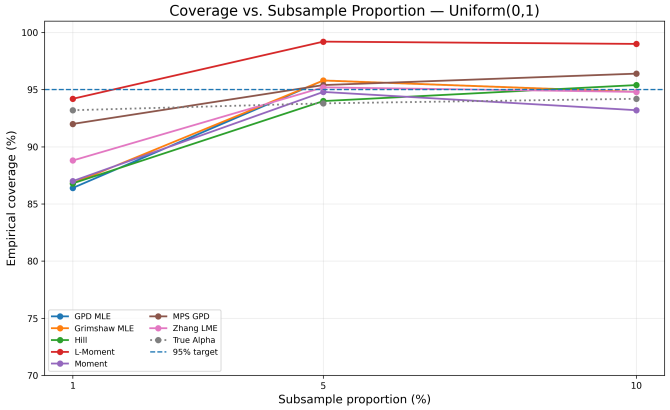


Fig. 1: Empirical coverage across subsample proportions for the Uniform$(0, 1)$ distribution.

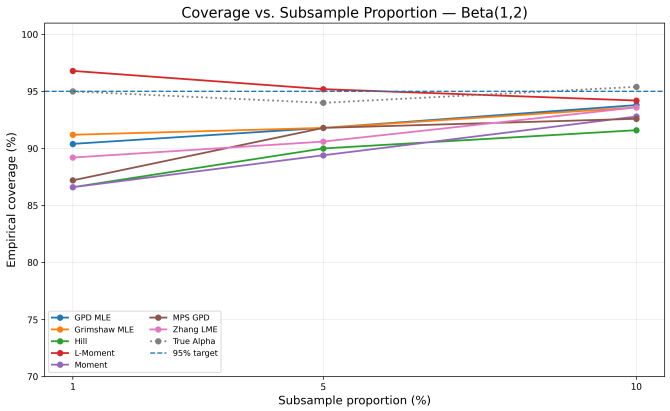


Fig. 2: Empirical coverage across subsample proportions for the Beta$(1, 2)$ distribution.

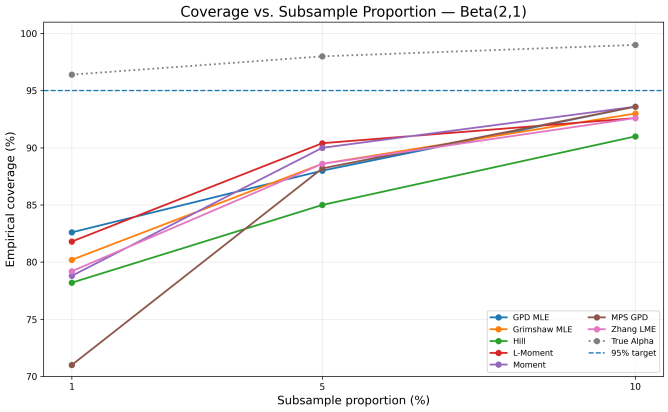


Fig. 3: Empirical coverage across subsample proportions for the Beta$(2, 1)$ distribution.

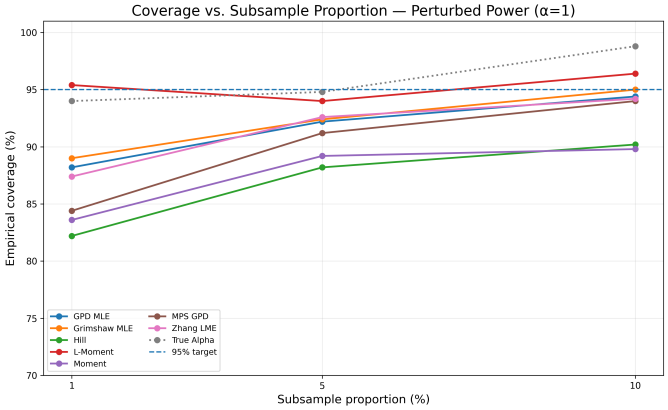


Fig. 4: Empirical coverage across subsample proportions for the Power distribution with $\alpha = 1$.

Across the four distributions, coverage generally improves as the subsample proportion increases. At the 1% subsample proportion, there is more variation in coverage between the estimators. As the subsample proportion increases, the estimators tend to move closer to the desired coverage and the differences between the estimators become smaller.

The effect of the subsample proportion is not the same across all distributions. We see that the Beta$(2, 1)$ distribution has greater undercoverage at the smaller subsample proportions. The true α gives a benchmark for the confidence interval when the error in estimating α is removed. Overall, the results suggest that increasing the amount of observed data improves the reliability of the confidence interval.

B. Average Confidence Interval Length

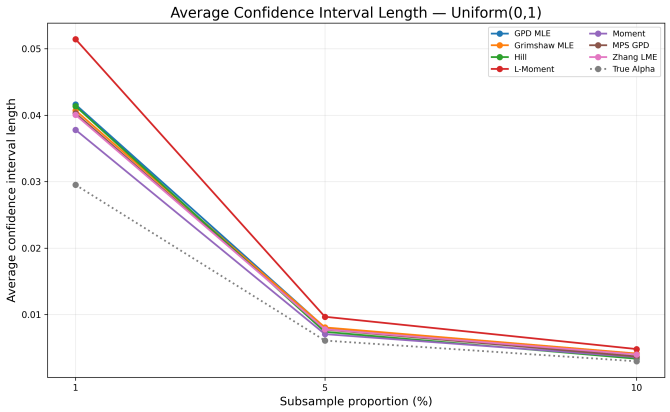


Fig. 5: Average confidence interval length across subsample proportions for the Uniform$(0, 1)$ distribution.

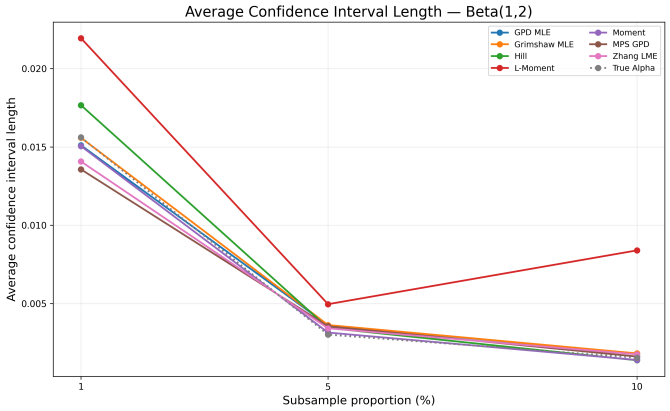


Fig. 6: Average confidence interval length across subsample proportions for the Beta$(1, 2)$ distribution.

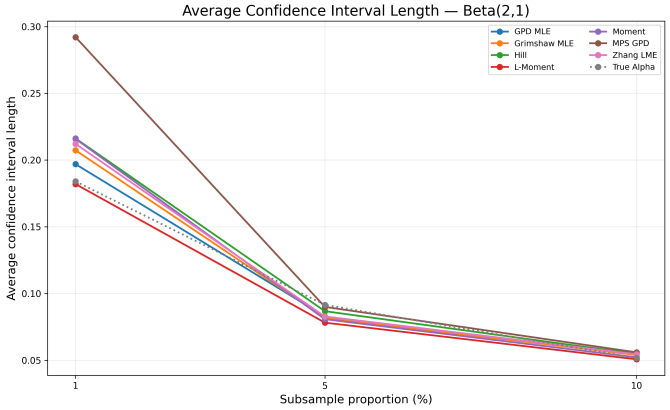


Fig. 7: Average confidence interval length across subsample proportions for the Beta$(2, 1)$ distribution.

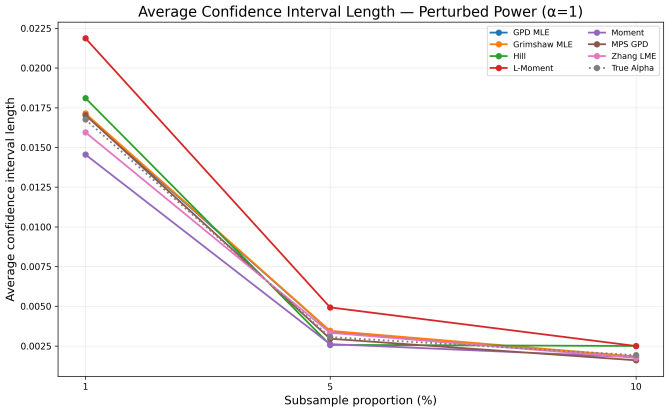


Fig. 8: Average confidence interval length across subsample proportions for the Power distribution with $\alpha = 1$.

Across all four distributions, the average confidence interval length decreases as the subsample proportion increases. The largest change occurs when jumping from a subsample proportion of 1% to 5%, while the jump from 5% to 10% is smaller. This suggests that observing a larger portion of the complete sample allows us to better estimate the minimum.

The confidence interval lengths also change between the distributions, as seen in Figures 5–8, with each distribution showing different behaviour. Overall, we see that increasing the subsample proportion across distributions allows us to estimate the minimum with greater precision.

C. *Effect and Selection of k*

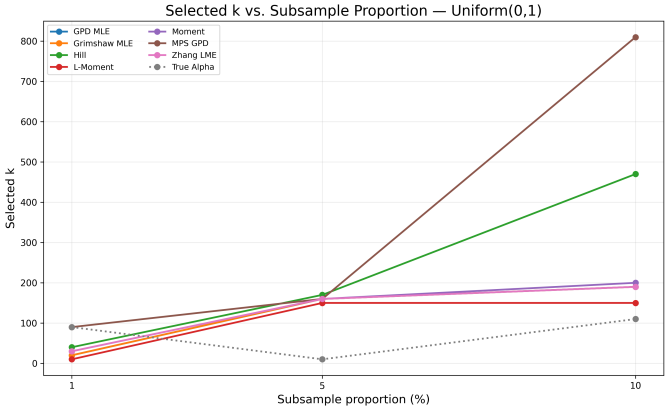


Fig. 9: Selected k across subsample proportions for the Uniform$(0, 1)$ distribution.

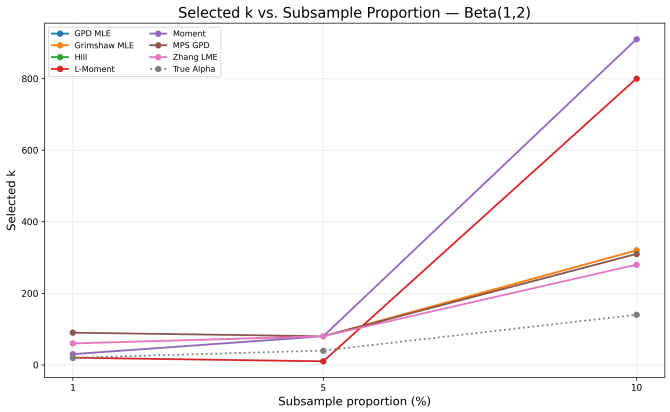


Fig. 10: Selected k across subsample proportions for the Beta$(1, 2)$ distribution.

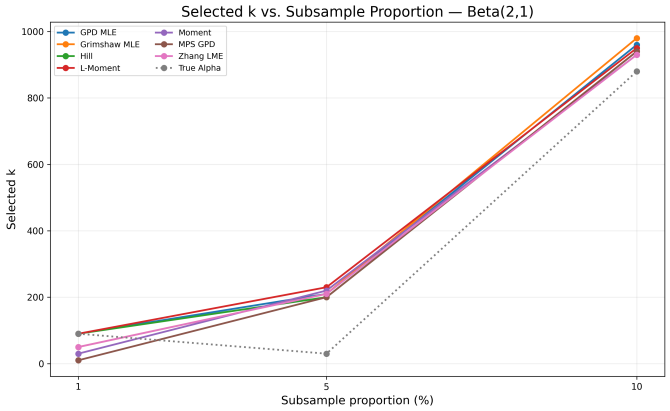


Fig. 11: Selected k across subsample proportions for the Beta$(2, 1)$ distribution.

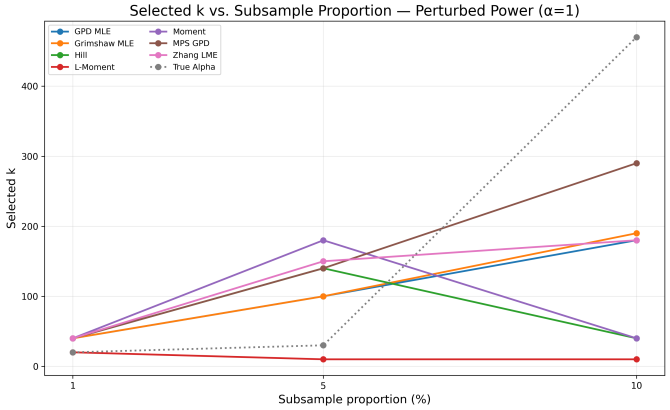


Fig. 12: Selected k across subsample proportions for the Power distribution with $\alpha = 1$.

The selected k varies depending on the distribution, estimator, and subsample proportion. The larger samples allow larger values of k to be selected because more observations are available, which may give the estimators more data to represent the lower tail. However, the pattern is not consistent for every estimator. Some perform better using smaller values of k, even when more observations are available.

The differences in selected k show that there is no single k value that works best for every method, and the appropriate k value depends on each estimator. The results also show the bias-variance tradeoff, since increasing k gives more observations, but values that are too large may begin to include observations that do not represent the lower tail.

D. Quality Score and Overall Estimator Performance

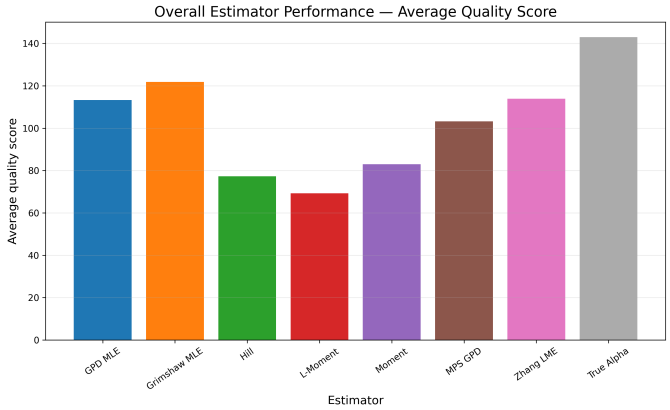


Fig. 13: Overall estimator performance based on the average quality score across the simulated distributions and subsample proportions.

Figure 13 shows a comparison of the overall performance of the estimators using the average quality score across the simulated distributions and subsample proportions. The quality score considers both coverage and average confidence interval length. This is important because coverage alone does not measure the precision of the confidence interval. An estimator may achieve desirable coverage close to 95% while producing wider intervals, while another estimator may produce shorter intervals but fail to contain the true minimum often enough.

The true-α benchmark produces the highest overall quality score. Among the methods that estimate α, Grimshaw MLE produces the highest average quality score, while GPD MLE and Zhang LME also show strong overall performance.

The performance of the true-α benchmark is important because it removes all the error associated with estimating the tail parameter. Its better overall quality score shows that estimating α contributes to the uncertainty in the final confidence interval. However, the relatively strong performance of several estimated-α methods shows that the procedure can still perform well when α is unknown and must be estimated from the observed subsample.

Overall, the previous results show that the performance of the minimum estimation procedure depends on the amount of

the complete sample that is observed, the estimator used to estimate α, and the amount of lower-tail information selected through k.

IV. Conclusion

The goal of this study was to evaluate and compare different tail-parameter estimation methods to determine which methods perform best when estimating the lower-tail parameter α. This was studied within a larger method for estimating the minimum of a large collection of values when only a subset of those values is observed. The motivation for this problem comes from Integrated Information Theory (IIT), where finding integrated information φ requires finding a minimum from a large number of possible partitions. As the size of the system increases, the number of possible partitions also increases, making it computationally infeasible to evaluate every possible partition. The overall motivation is therefore to determine whether we can instead observe only a subset of these values and use Extreme Value Theory (EVT) to estimate the minimum of the complete set.

Since the method is used to estimate a minimum, the most important information comes from the observations closest to the lower endpoint. The method therefore focuses on the lower tail rather than attempting to model the entire distribution. Since many EVT estimators are made for heavy upper tails, the lower-tail observations are transformed into an upper-tail problem. This allows different EVT estimators to be used to estimate the lower-tail parameter α, which is then used in the lower-tail quantile model to estimate the minimum of the complete sample.

An important part of the experiment is the estimation of α. Different EVT estimators may produce different estimates of α even when they are given the same observations. The simulation therefore compares multiple estimators under the same conditions to determine how the choice of estimator affects the final confidence interval. The true α is also used as a benchmark where no estimation of α is needed, allowing us to compare the estimated-α methods to the performance of the procedure when the error from estimating α is removed.

The simulation results show that the estimator used to estimate α can affect the performance of the final confidence interval. The true-α benchmark produced the highest overall quality score, showing that uncertainty from estimating α contributes to the uncertainty in the final confidence interval. Among the methods that estimate α, Grimshaw MLE produced the highest average quality score. GPD MLE and Zhang LME also showed strong overall performance. These results suggest that Grimshaw MLE, GPD MLE, and Zhang LME are the strongest options considered in this study when α must be estimated from the observed subsample.

The value of k also affects estimator performance because it controls the amount of information considered to belong to the tail. The results show that there is no single value of k that performs best for every estimator. Different estimators may perform better using different amounts of tail information, supporting the use of an estimator-specific value of k. The amount of the complete sample that is observed is also important, as increasing the subsample proportion generally improves coverage and decreases confidence interval length.

These results also show that the lower tail of an observed subsample can provide information about values that have not been observed. This is important for IIT because the complete set of partitions may become too large to evaluate. If the partition values follow lower-tail behaviour that can be modeled using this method, then only a subset of the partitions may need to be observed. The lower tail of this subset could then be used to estimate the minimum of the complete collection. This would not remove the problem of finding the minimum completely, but it may reduce the amount of computation needed compared to evaluating every possible partition.

There are still limitations to the current experiment. The simulation uses controlled distributions where the lower-tail behaviour and true α are known. Actual IIT partition values may not follow the same behaviour as the distributions used in this experiment. The method also depends on the selection of k and how well the lower-tail model represents the actual observations near the endpoint. Because of this, the results from the simulation do not guarantee that the same performance will occur when the method is applied to actual IIT partition values.

Future work should test the method using actual partition values produced from IIT calculations. One way to test this would be to use smaller systems where evaluating every possible partition is still possible. The actual minimum could first be found by evaluating the complete set, and then only a random subset of the partitions could be given to the estimation procedure. This would allow us to compare the estimated confidence interval to the actual minimum and determine whether the method works outside of the simulated distributions. Future work could also test more distributions, different subsample sizes, different methods of selecting k, and other methods for estimating α.

Overall, this study shows that the choice of tail-parameter estimator is an important part of the minimum-estimation procedure. Grimshaw MLE produced the strongest overall performance among the methods that estimate α, while GPD MLE and Zhang LME also performed well. The results also show that estimator performance depends on the amount of observed data and the amount of lower-tail information selected through k. Although more testing is needed before this can be applied directly to IIT, these results identify promising tail-parameter estimation methods for continuing to study whether a minimum can be estimated without evaluating the complete set.

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